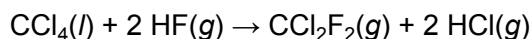
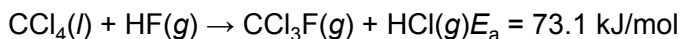


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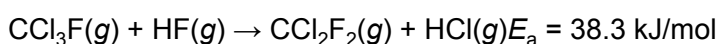


Reaction 1

The formation of Freon through Reaction 1 occurs via a two-step reaction mechanism, with each step catalyzed by chromium(III) fluoride (CrF_3).



Reaction 2



Reaction 3

During a refrigeration cycle, gaseous Freon is pressurized in a compressor and liquified. The liquified Freon releases its heat into the surroundings as it travels to a throttling valve, where it expands. As the liquid Freon expands, it rapidly becomes a gas while absorbing heat and cooling the surroundings. The cold gaseous Freon is recovered and another cycle begins. The relationship between pressure and enthalpy of Freon during a refrigeration cycle is shown in the phase diagram in Figure 1.

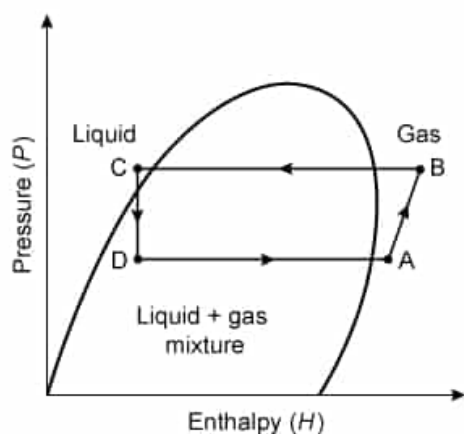


Figure 1 A pressure-enthalpy (P - H) phase diagram of Freon during a refrigeration cycle (Note: The curve represents the boundary between the liquid and gas phases.)

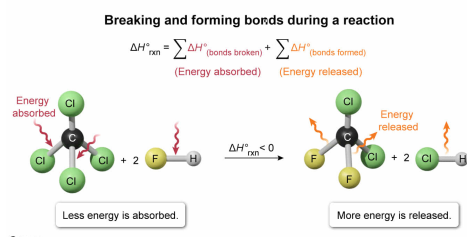
Which of the following statements correctly describes the energy associated with the breaking and forming of bonds during Reaction 1?

- ☐ A. The energy absorbed when breaking the bonds of the reactants is greater than the energy released when forming the bonds of the products. [14%]
- ☒ B. The energy absorbed when breaking the bonds of the reactants is less than the energy released when forming the bonds of the products. [43%]
- ☐ C. The energy released when breaking the bonds of the reactants is greater than the energy absorbed when forming the bonds of the products. [33%]
- ☐ D. The energy released when breaking the bonds of the reactants is less than the energy absorbed when forming the bonds of the products. [7%]

Correct

43% answered correctly

Explanation:



Bond enthalpy ΔH° (bond dissociation energy) is the amount of energy needed to break 1 mole of a bond between two atoms in the gas phase at a temperature of 298 K. **Forming a bond** is an **exothermic** process ($\Delta H^\circ_{\text{bonds formed}} < 0$) whereas **breaking a bond** is an **endothermic** process ($\Delta H^\circ_{\text{bonds broken}} > 0$).

Consequently, the enthalpies associated with breaking or forming a given type of bond are equal in magnitude but opposite in sign.

A chemical reaction can be viewed as a series of steps in which reactant bonds are broken and product bonds are formed. The overall **change in enthalpy of a reaction** $\Delta H^\circ_{\text{rxn}}$ measures the **heat released or consumed** (ie, $\Delta H^\circ_{\text{rxn}} < 0$ or $\Delta H^\circ_{\text{rxn}} > 0$, respectively) by 1 **mole of reaction**. Hess's law states that if a reaction occurs in a series of **intermediate steps**, $\Delta H^\circ_{\text{rxn}}$ is equal to the **sum** of the **enthalpy changes** for each step. Accordingly, $\Delta H^\circ_{\text{rxn}}$ can be calculated by:

$$\Delta H^\circ_{\text{rxn}} = \sum \Delta H^\circ_{\text{(bonds broken)}} + \sum \Delta H^\circ_{\text{(bonds formed)}}$$

If $\Delta H^\circ_{\text{rxn}} > 0$ (ie, an endothermic reaction), then $\sum \Delta H^\circ_{\text{(bonds broken)}} > \sum \Delta H^\circ_{\text{(bonds formed)}}$ (ie, more energy was absorbed than released) and if $\Delta H^\circ_{\text{rxn}} < 0$ (ie, an exothermic reaction), then $\sum \Delta H^\circ_{\text{(bonds broken)}} < \sum \Delta H^\circ_{\text{(bonds formed)}}$ (ie, more energy was released than absorbed).

According to the passage, the formation of Freon from Reaction 1 is an exothermic process ($\Delta H^\circ_{\text{rxn}} < 0$).

Therefore, the amount of energy absorbed during bond breakage is less than the energy released during bond formation.

(Choices A, C, and D) Breaking a bond requires energy absorption ($\Delta H^\circ > 0$) whereas forming a bond releases energy ($\Delta H^\circ < 0$). Because $\Delta H^\circ_{\text{rxn}} < 0$, less energy was absorbed than released.

Educational objective:

Bond enthalpy is the energy needed to break 1 mole of a given type of bond between two atoms in the gas phase at 298 K. According to Hess' law, a reaction can be assessed as a series of endothermic bond-breaking and exothermic bond-forming steps, and the heat of the reaction is equal to the sum of the enthalpy changes for each step.

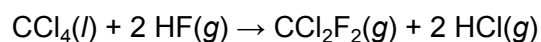
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Subject:
General Chemistry

QID: 404046

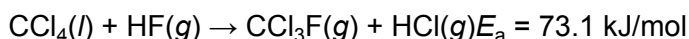
System:
Thermodynamics, Kinetics & Gas Laws

Health care facilities such as hospitals employ refrigeration to safeguard temperature-sensitive materials such as vaccines and drugs from developing life-threatening bacteria and toxins. Some refrigeration systems utilize Freon (CCl_2F_2) as a refrigerant, which is produced through an exothermic reaction between $\text{CCl}_4(l)$ and $\text{HF}(g)$.

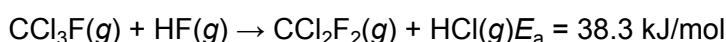


Reaction 1

The formation of Freon through Reaction 1 occurs via a two-step reaction mechanism, with each step catalyzed by chromium(III) fluoride (CrF_3).



Reaction 2



Reaction 3

During a refrigeration cycle, gaseous Freon is pressurized in a compressor and liquified. The liquified Freon releases its heat into the surroundings as it travels to a throttling valve, where it expands. As the liquid Freon expands, it rapidly becomes a gas while absorbing heat and cooling the surroundings. The cold gaseous Freon is recovered and another cycle begins. The relationship between pressure and enthalpy of Freon during a refrigeration cycle is shown in the phase diagram in Figure 1.

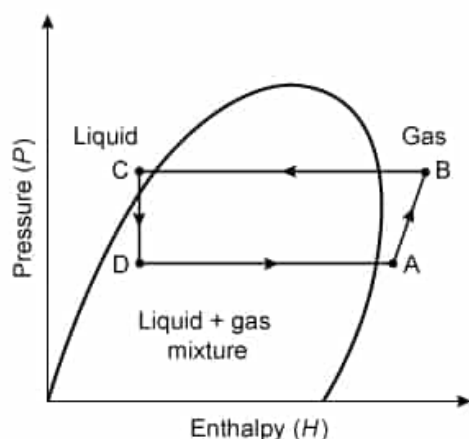


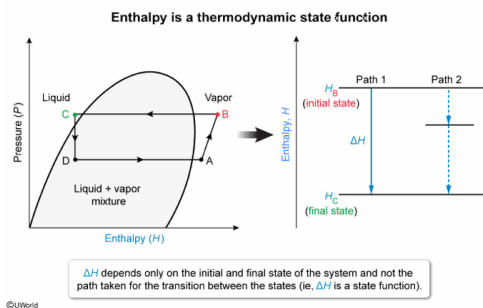
Figure 1 A pressure-enthalpy (P - H) phase diagram of Freon during a refrigeration cycle (Note: The curve represents the boundary between the liquid and gas phases.)

During the refrigeration cycle in Figure 1, does the change in enthalpy from Point B to Point C depend on the path taken to achieve the final state?

- ☐ A. Yes, because enthalpy is a state function. [15%]
- ☐ B. Yes, because enthalpy is not a state function. [5%]
- ✓ ☒ C. No, because enthalpy is a state function. [66%]
- ☐ D. No, because enthalpy is not a state function. [11%]

Correct
66% answered correctly

Explanation:



A **state function** describes an intrinsic property of a thermodynamic system whose value depends only on the **state** of the system. As such, a state function is **independent of the path taken** (ie, the process by which the state was achieved) to reach a specific value.

For a thermodynamic system held at constant pressure, the change in total energy is described by the change in enthalpy ΔH . The value of ΔH depends **only** on the initial state (H_{initial}) and the final state (H_{final}) of the system. Consequently, enthalpy is a state function. Therefore, as the Freon undergoes a thermodynamic transition from Point B (ie, the initial state) to Point C (ie, the final state) at constant pressure, the resulting change in enthalpy **does not depend** on the path taken to achieve the final state.

(Choices A, B, and D) Because enthalpy is a state function, the change in enthalpy depends only on the initial and final states of the system, not the path taken.

Educational objective:

A state function depends only on the state of a system, irrespective of the path taken to achieve a given state.

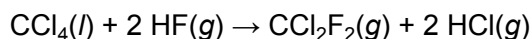
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Subject:
General Chemistry

QID: 404047

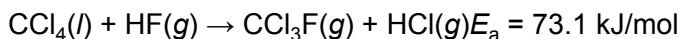
System:
Thermodynamics, Kinetics & Gas Laws

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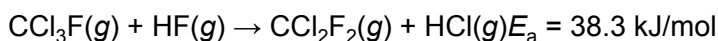


Reaction 1

The formation of Freon through Reaction 1 occurs via a two-step reaction mechanism, with each step catalyzed by chromium(III) fluoride (CrF_3).



Reaction 2



Reaction 3

During a refrigeration cycle, gaseous Freon is pressurized in a compressor and liquified. The liquified Freon releases its heat into the surroundings as it travels to a throttling valve, where it expands. As the liquid Freon expands, it rapidly becomes a gas while absorbing heat and cooling the surroundings. The cold gaseous Freon is recovered and another cycle begins. The relationship between pressure and enthalpy of Freon during a refrigeration cycle is shown in the phase diagram in Figure 1.

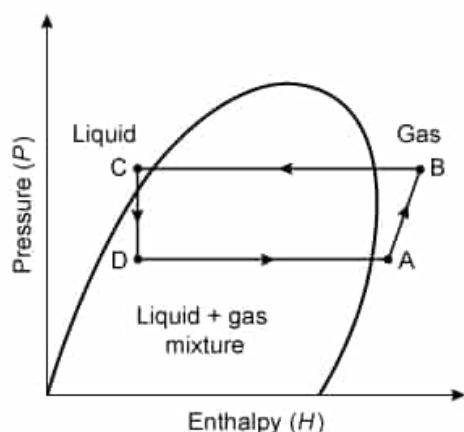
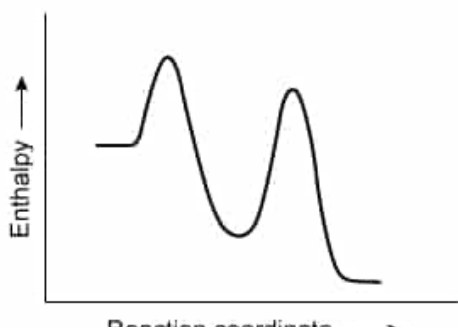


Figure 1 A pressure-enthalpy (P - H) phase diagram of Freon during a refrigeration cycle (Note: The curve represents the boundary between the liquid and gas phases.)

Which of the following diagrams illustrates the energy profile for the formation of Freon in Reaction 1?

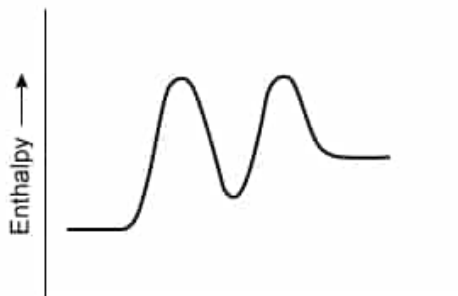
☐ A.



Reaction coordinate →

[7%]

☐ B.



Reaction coordinate →

[10%]

✓ ☒ C.



Reaction coordinate →

[80%]

☐ D.



Reaction coordinate →

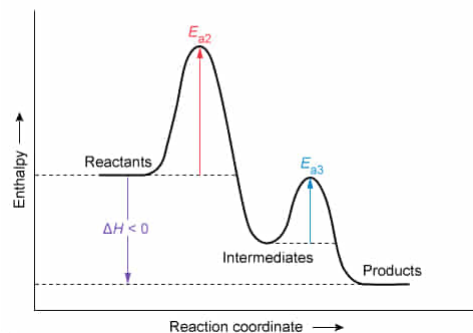
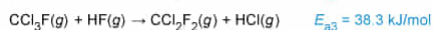
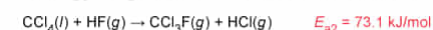
[2%]

Correct

80% answered correctly

Explanation:

Energy diagram for the formation of Freon



The change in the total energy of a system is indicated by its **change in enthalpy** ΔH . The energy changes that occur during the progress of a reaction may sometimes be represented on a diagram that plots the **enthalpy** H (y-axis) versus the **reaction coordinate** (x-axis), which indicates the reaction progress. ΔH is evaluated as:

$$\Delta H = H_{\text{products}} - H_{\text{reactants}}$$

For an **exothermic reaction** ΔH is negative (ie, $H_{\text{products}} < H_{\text{reactants}}$), and for an **endothermic reaction** ΔH is positive (ie, $H_{\text{products}} > H_{\text{reactants}}$).

The **activation energy** E_a (ie, the minimum amount of energy required to initiate a reaction) is the difference between the energy of the **transition state** and the energy of the reactants. Because E_a is the energy required by the reactants to form a higher energy transition state, E_a is **always positive**.

According to the passage, the E_a for Reaction 2 ($E_{a2} = 73.1$ kJ/mol) is greater than the E_a for Reaction 3 ($E_{a3} = 38.3$ kJ/mol). In addition, ΔH is a negative value for the overall reaction (Reaction 1), so the energy of the reactants is greater than the energy of the products. Therefore, the energy diagram shown in **Choice C** best depicts the energetics for the formation of Freon.

(Choice A) This energy diagram incorrectly shows $E_{a2} < E_{a3}$, rather than $E_{a2} > E_{a3}$.

(Choice B) Although this energy diagram correctly shows $E_{a2} > E_{a3}$, the energy of the reactants is incorrectly shown as less than that of the products. Because the overall reaction is *exothermic*, the energy of the reactants is *greater* than that of the products.

(Choice D) E_{a2} and E_{a3} are incorrectly shown as inverted peaks. The activation energy for a reaction is always seen as a peak on an energy diagram.

Educational objective:

The change in enthalpy ΔH evaluates the total energy of a system. For an exothermic reaction ($\Delta H < 0$), the energy of the reactants is greater than the energy of the products. The difference in energy between the transition state and the reactants is the activation energy, which is always a positive value.

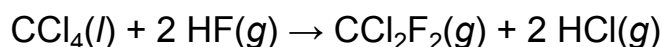
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Subject:
General Chemistry

QID: 404048

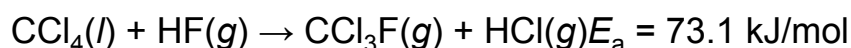
System:
Thermodynamics, Kinetics & Gas Laws

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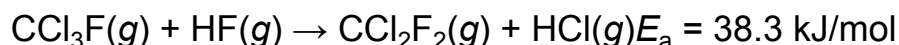


Reaction 1

The formation of Freon through Reaction 1 occurs via a two-step reaction mechanism, with each step catalyzed by chromium(III) fluoride (CrF_3).



Reaction 2



Reaction 3

During a refrigeration cycle, gaseous Freon is pressurized in a compressor and liquified. The liquified Freon releases its heat into the surroundings as it travels to a throttling valve, where it expands. As the liquid Freon expands, it rapidly becomes a gas while absorbing heat and cooling the surroundings. The cold gaseous Freon is recovered and another cycle begins. The relationship between pressure and enthalpy of Freon during a refrigeration cycle is shown in the phase diagram in Figure 1.

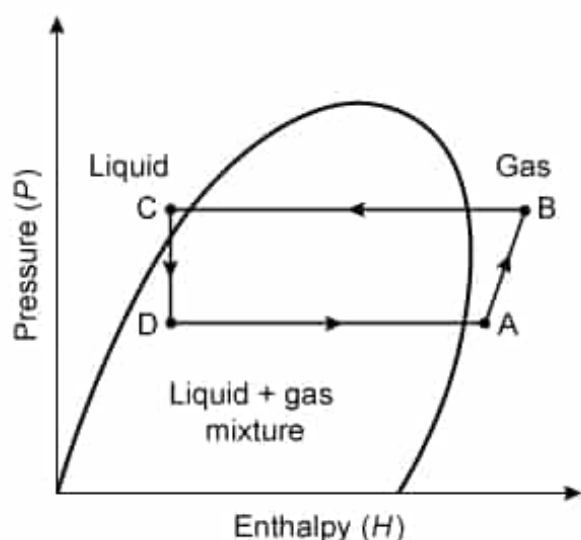


Figure 1 A pressure-enthalpy (P - H) phase diagram of Freon during a refrigeration cycle (Note: The curve represents the boundary between the liquid and gas phases.)

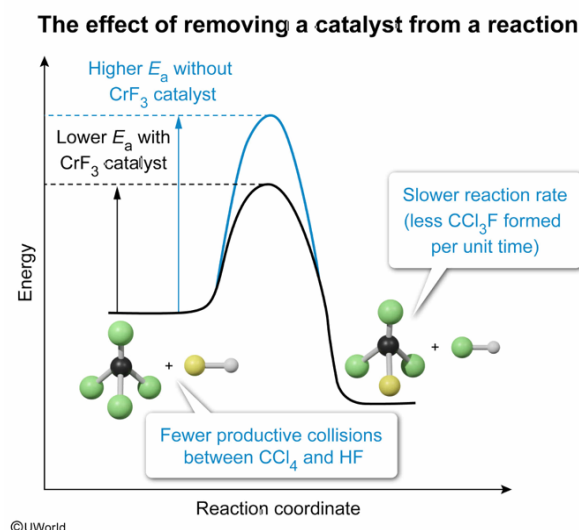
Which of the following will occur if CrF_3 is NOT added to Reaction 2?

- I. The rate of the reaction will increase.
 - II. Fewer productive collisions between $\text{CCl}_4(l)$ and $\text{HF}(g)$ will occur per unit time.
 - III. Less $\text{CCl}_3\text{F}(g)$ will be produced per unit time.
 - IV. The total amount of $\text{CCl}_4(l)$ consumed will decrease.
- ☐ A. I and II only [3%]
- ✓ ☒ B. II and III only [70%]
- ☐ C. I, II, and IV only [2%]
- ☐ D. II, III, and IV only [22%]

Correct

70% answered correctly

Explanation:



The **rate of a reaction** indicates how quickly the reactant molecules are consumed and the product molecules are produced (ie, the change in the species' concentrations per unit time). A **catalyst** is a substance that **increases the reaction rate** but is **not consumed** in the overall reaction. Catalysts work by providing an alternative reaction mechanism that forms a more stable (ie, **less energetic**) **transition state** and thereby **lowering** the **activation energy E_a** of the **rate-determining step** (ie, less energy is required to initiate the reaction). This in turn **increases** the number of **productive collisions** between reactant molecules, which results in a faster reaction rate.

The passage states that Reaction 2 is catalyzed by CrF_3 . As such, CrF_3 lowers the E_a of the reaction. If CrF_3 is not present during the reaction, the E_a of the reaction will be higher, yielding fewer productive collisions between $\text{CCl}_4(l)$ and $\text{HF}(g)$ **(Number II)**. Consequently, less $\text{CCl}_3\text{F}(g)$ (and $\text{HCl}(g)$) will be produced per unit time **(Number III)**.

(Number I) A catalyst increases the rate of reaction. If a catalyst is *not* present, the rate of reaction *decreases*.

(Number IV) Not having a catalyst present during Reaction 2 decreases the amount of $\text{CCl}_4(l)$ consumed *per unit time* (ie, how *fast* the reactant is consumed), not the *total amount* of $\text{CCl}_4(l)$ consumed.

Educational objective:

Catalysts decrease the activation energy of the rate-determining step of a reaction, thereby increasing the reaction rate (ie, the change in species concentration per unit time).

Time spent:0 sec

Subject:

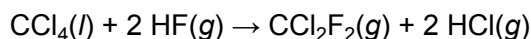
General Chemistry

QID:404049

System:

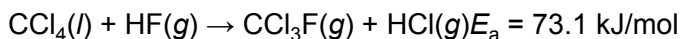
Thermodynamics, Kinetics & Gas Laws

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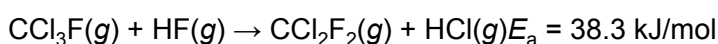


Reaction 1

The formation of Freon through Reaction 1 occurs via a two-step reaction mechanism, with each step catalyzed by chromium(III) fluoride (CrF_3).



Reaction 2



Reaction 3

During a refrigeration cycle, gaseous Freon is pressurized in a compressor and liquified. The liquified Freon releases its heat into the surroundings as it travels to a throttling valve, where it expands. As the liquid Freon expands, it rapidly becomes a gas while absorbing heat and cooling the surroundings. The cold gaseous Freon is recovered and another cycle begins. The relationship between pressure and enthalpy of Freon during a refrigeration cycle is shown in the phase diagram in Figure 1.

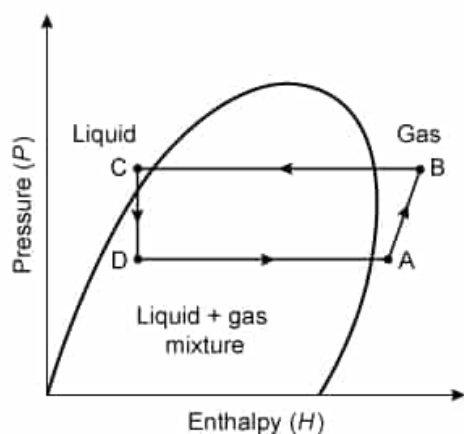


Figure 1 A pressure-enthalpy (P - H) phase diagram of Freon during a refrigeration cycle (Note: The curve represents the boundary between the liquid and gas phases.)

The arrows proceeding from Point B to Point C and from Point D to Point A on the refrigeration cycle in Figure 1 represent which phase changes, respectively?

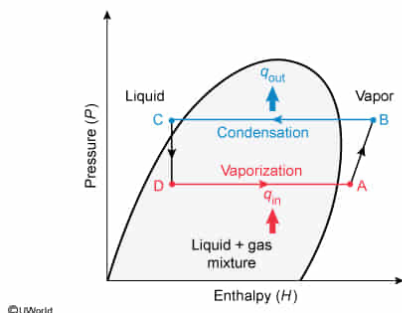
- ☒ A. Condensation; vaporization [95%]
- ☐ B. Vaporization; condensation [2%]
- ☐ C. Sublimation; condensation [1%]
- ☐ D. Vaporization; sublimation

Correct

95% answered correctly

Explanation:

Phase changes of Freon during a refrigeration cycle



A substance (ie, matter) can exist in one of three common phases (ie, states): a gas, liquid, or solid. When enough energy (transferred as heat) is either absorbed or released by a substance, a physical process can occur in which the substance transitions from one phase to another (ie, a **phase change**). There are six types of phase changes that can occur between the solid, liquid, and gas states:

- Matter can transition from a gas to a liquid (**condensation**) or from a liquid to a gas (**vaporization**).
- Matter can transition from a liquid to a solid (**solidification**) or from a solid to a liquid (**melting**).
- Matter can transition from a solid to a gas (**sublimation**) or from a gas to a solid (**deposition**).

As described in the passage, a pressure-enthalpy (P - H) phase diagram represents the relationship between pressure and enthalpy of a refrigerant. On the P - H diagram, the region within the curve represents a mixture of both gas- and liquid-phase refrigerant. To the right of the curve, the refrigerant is in the gas phase and to the left of the curve, the refrigerant is in the liquid phase.

In the refrigeration cycle shown on the P - H diagram in Figure 1, Freon is in the gas phase at Point B and transitions to the liquid phase at Point C while releasing heat. Therefore, **condensation** occurs from Point B to Point C. As the refrigeration cycle continues, a mixture of gas and liquid phase Freon is present at Point D, and the liquid Freon transitions back to the gas phase at Point A as heat is absorbed. Therefore, **vaporization** occurs from Point D to Point A.

(Choice B) Freon *condenses* from Point B to Point C and *vaporizes* from Point D to Point A.

(Choices C and D) Sublimation is the direct conversion of a substance in the solid phase to the gas phase. According to Figure 1, Freon is in only the gas or liquid phase during a refrigeration cycle.

Educational objective:

Condensation is the transition of a substance from the gas phase to the liquid phase. Vaporization is the reverse of condensation.

Time spent: 0 sec

Subject:
General Chemistry

QID: 404050

System:
Thermodynamics, Kinetics & Gas Laws